Oblique asymptotes and range of rational graphs

Key Points

For the rational function $f(x) = \frac{N(x)}{D(x)}$

If $f(x) \to \pm\infty$ we can find how the function tends to infinity along its oblique (slanted) asymptote/limiting function by using partial fractions/polynomial division

To find the range of a rational function find for which $k \in \mathbb{R}$ there is a solution to $\frac{N(x)}{D(x)} = k$

(If quadratic use determinant to determine when solutions)

Basic Skills

Q1. Without solving the equation $x^2 + 5x + 3 = 0$ determine how many real solutions it will have

Q2. For which constants k does the equation $x^2 - 2kx + 4 = 0$ have no real solutions

Q3. Find the constants such that $\frac{x^2+3}{x-1} = Ax + B + \frac{c}{x-1}$

Q4. Use polynomial division to express the rational function $f(x) = \frac{3x^2-2x+10}{2x+1}$ as a linear equation and a remainder

Q5. State what happens to $y = \frac{6x}{8-4x}$ as $x \to \infty$

Q6. Sketch the graph of $y = \frac{x}{x^2-4x-12}$

Competency Skills

Q7. For the function $f(x) = \frac{x^2-2x+3}{x+1}$

a) Express $f(x) = Ax + B + \frac{C}{x+1}$ for some constants A, B, C

b) Hence state the oblique asymptote for the function $f(x)$ and draw the graph of $y = f(x)$

Q8. By solving the equation $x^2 + 5x - 1 = c(x^2 - 2x + 1)$ determine the range of the graph $y = \frac{x^2+5x-1}{x^2-2x+1}$

Q9. Sketch the graph of $y = \frac{x^3+6x}{x^2-2}$ indicating the oblique asymptote

Q10. Sketch the graph of $y = \frac{x^2-7x+10}{x^2+10x+21}$ indicating the range

Q11. Show that $1 \leq \frac{9x^2+8x+3}{x^2+1} \leq 11$

Q12. Sketch the graph of $y = \frac{x^3}{x^2-x-20}$

Advanced Skills

Q13. Explain under what conditions a rational function goes to infinity (or negative infinity) as $x \to \pm\infty$

Q14. For the graph of $y = \frac{x}{(x-a)^2}$ for any real constant a state the special behaviour around the vertical asymptote and comment on why it is special

Q15. By sketching an appropriate graph and working out the range state the local maxima and minima of the function $f(x) = \frac{x+1}{(x+2)(x+3)}$ without differentiating

Q16. For what value of a does the equation $y = \frac{x^2+ax+10}{x-5}$ not have any oblique asymptotes

Q17. Explain in detail what happens to the function $f(x) = \frac{x^4-5x+2}{x^2-4}$ as $x \to \infty$

Extension

Q18. Asymptotic behaviour (limits of infinity behaviour) models how a function tends to infinity and how fast. Research Big O notation and decide if e^x goes to infinity quicker than any polynomial